Web-based materials/chemistry simulation platform

Materials Square
Class Curriculum

Catalyst

[Lab.3] Oxygen Reduction Reaction (ORR)

Learning Objectives:

→ Verify the Oxygen Reduction Reaction (ORR) process on the electrocatalysts using density functional theory (DFT) calculations.

Materials: Platinum (111) surfaces

MatSQ Credit: ~ 172 CPU hour ($ 43.0)

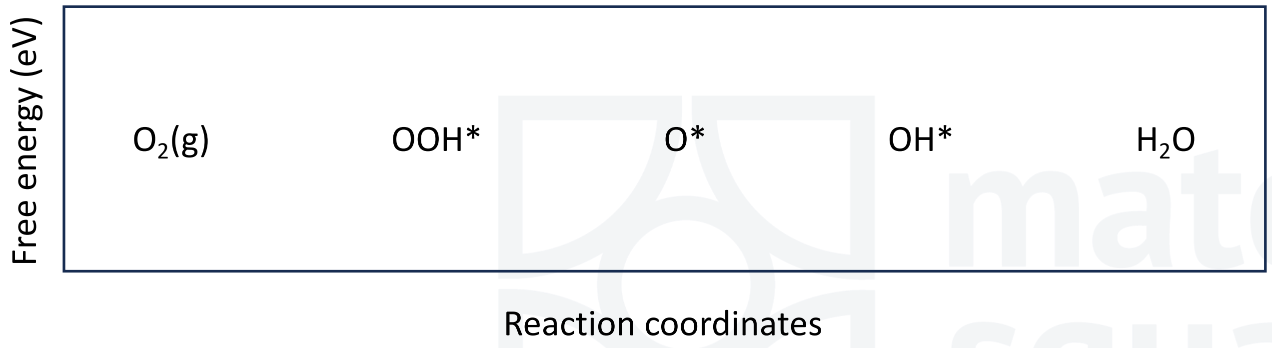
[Lab.3] Oxygen Reduction Reaction (ORR)

Pt(111) surfaces

1. Build a Pt(111) surface structure
2. Optimize the atomic structure of Pt(111) surface
3. Build adsorption structures
4. Optimize atomic structures (adsorption structures)
5. Analyze optimized adsorption structures
6. Build molecule structures
7. Optimize molecule structures using High Throughput
8. Calculate adsorption energies
9. Calculate free-energy diagram

MatSQ Credit: ~ 172 CPU hour ($ 43.0)

ORR Free energy diagram



$$\Delta E_{\mathrm{OOH}^*} = E(\mathrm{OOH}^*/slab) - E(slab) - \left\{2E(\mathrm{H_2O}) - \frac{3}{2}E(\mathrm{H_2})\right\}$$

$$\Delta E_{\mathrm{O}^*} = E(\mathrm{O}^*/slab) - E(slab) - \{E(\mathrm{H_2O}) - E(\mathrm{H_2})\}$$

$$\Delta E_{\mathrm{OH}^*} = E(\mathrm{OH}^*/slab) - E(slab) - \left\{E(\mathrm{H_2O}) - \frac{1}{2}E(\mathrm{H_2})\right\}$$

List of required calculations

- Molecules : H_2, H_2O,
- Substrate: Pt(111)
- Adsorption structures:O_2/Pt(111), OOH/Pt(111), O/Pt(111), OH/Pt(111)

[Lab.3] Oxygen Reduction Reaction (ORR)

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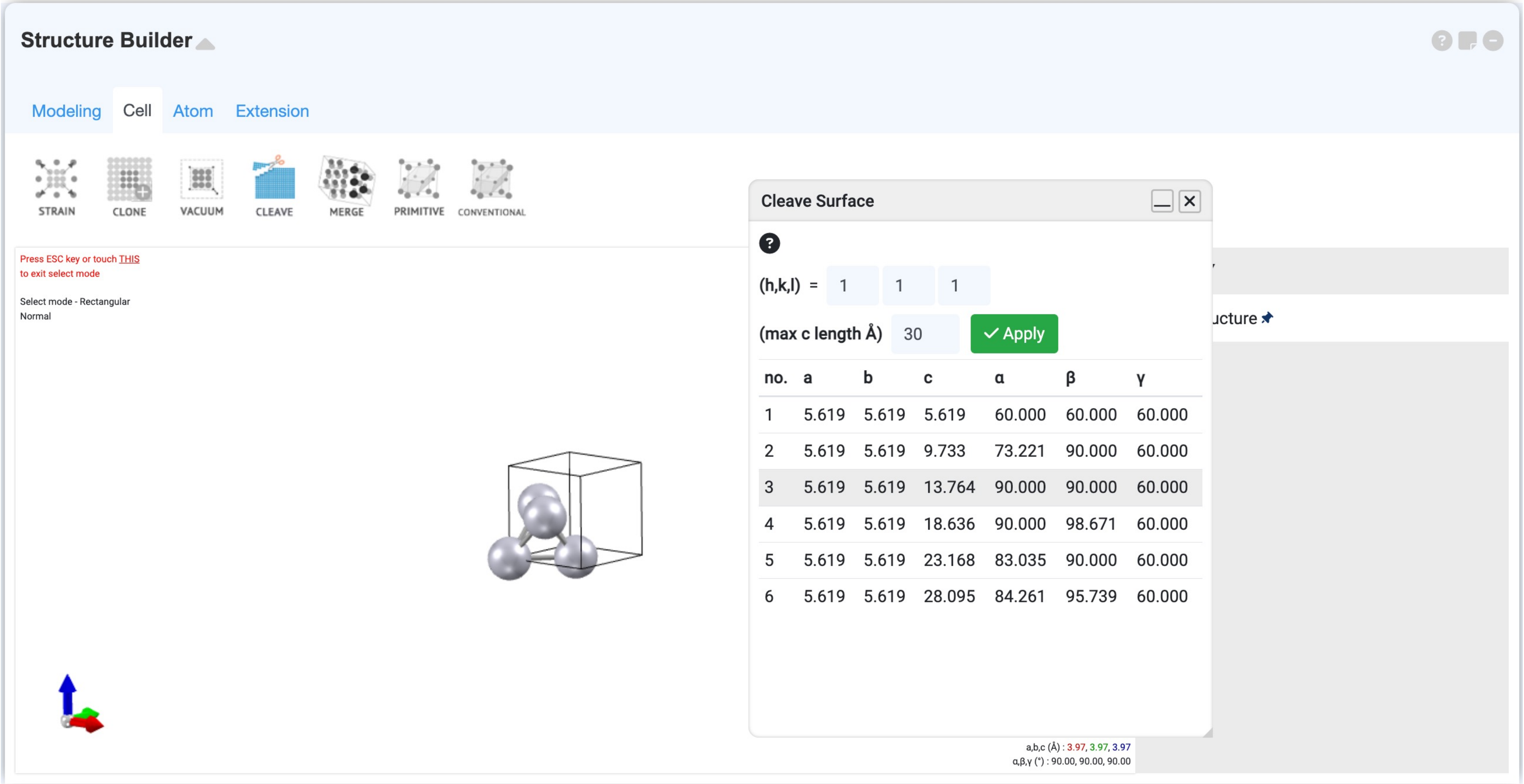
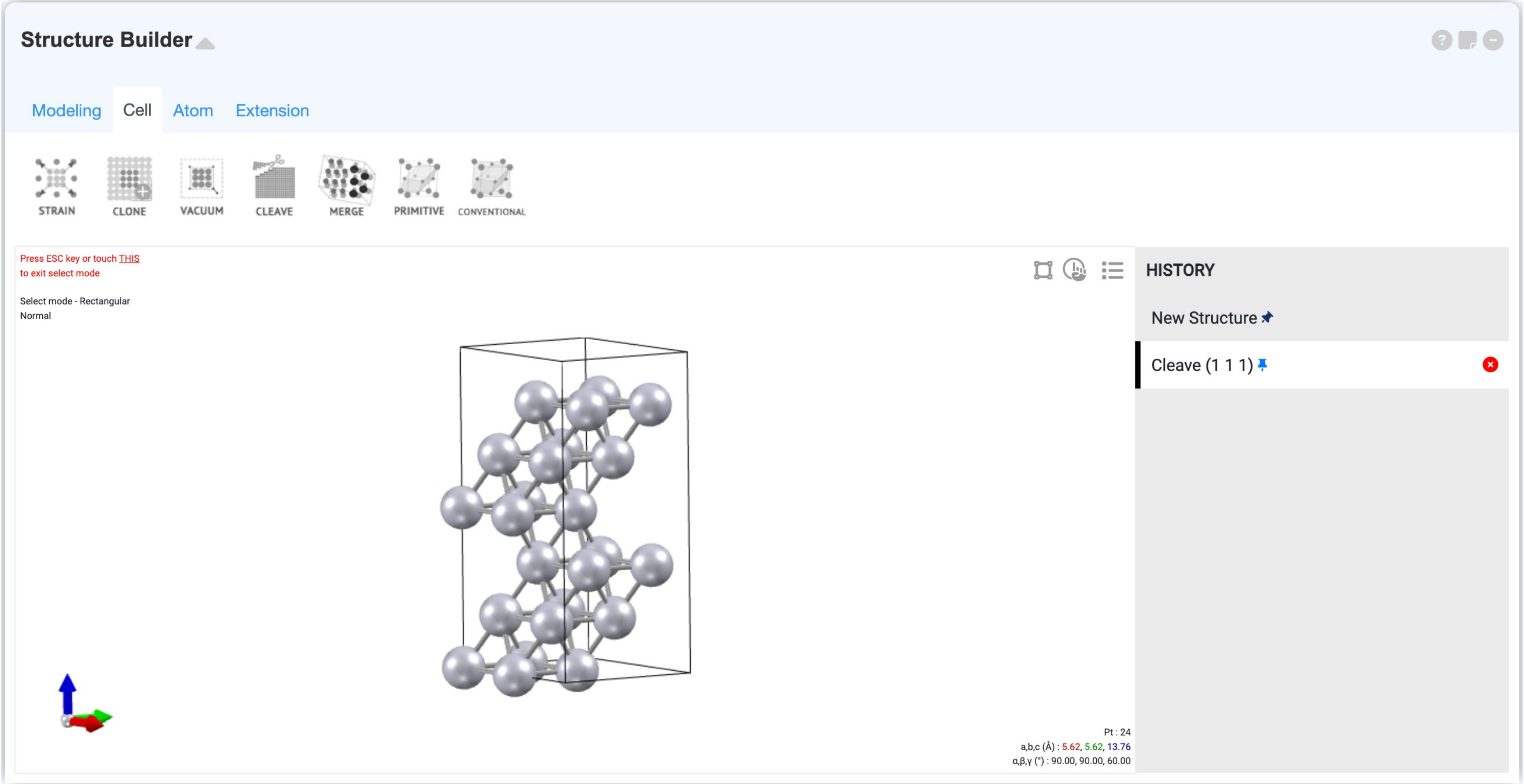
1. Build a Pt(111) surface structure



no.	a	b	c	α	β	γ
1	5.619	5.619	5.619	60.000	60.000	60.000
2	5.619	5.619	9.733	73.221	90.000	60.000
3	5.619	5.619	13.764	90.000	90.000	60.000
4	5.619	5.619	18.636	90.000	98.671	60.000
5	5.619	5.619	23.168	83.035	90.000	60.000
6	5.619	5.619	28.095	84.261	95.739	60.000

[Lab.3] Oxygen Reduction Reaction (ORR)

1. Build a Pt(111) surface structure



[Lab.3] Oxygen Reduction Reaction (ORR)

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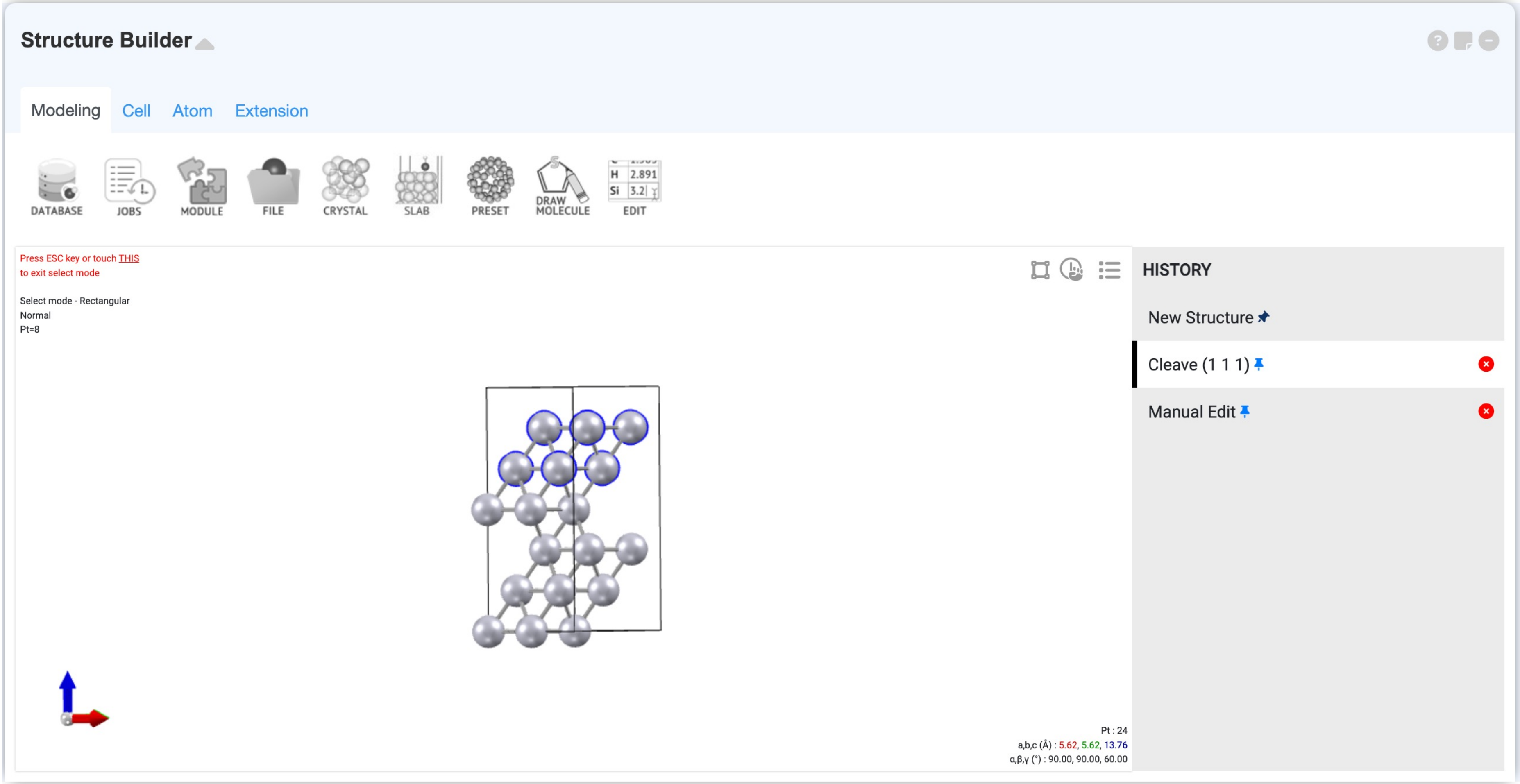

[Lab.3] Oxygen Reduction Reaction (ORR)

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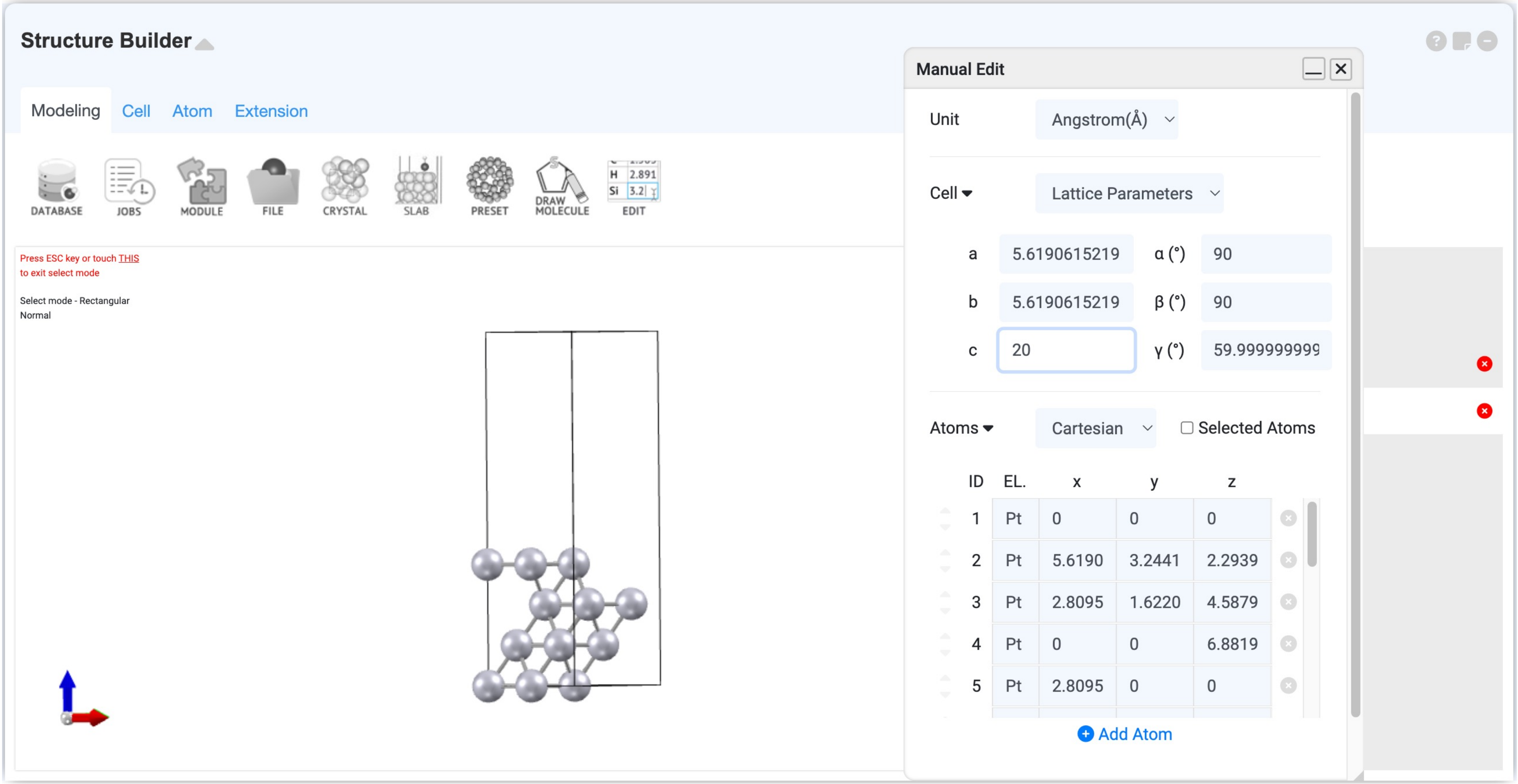
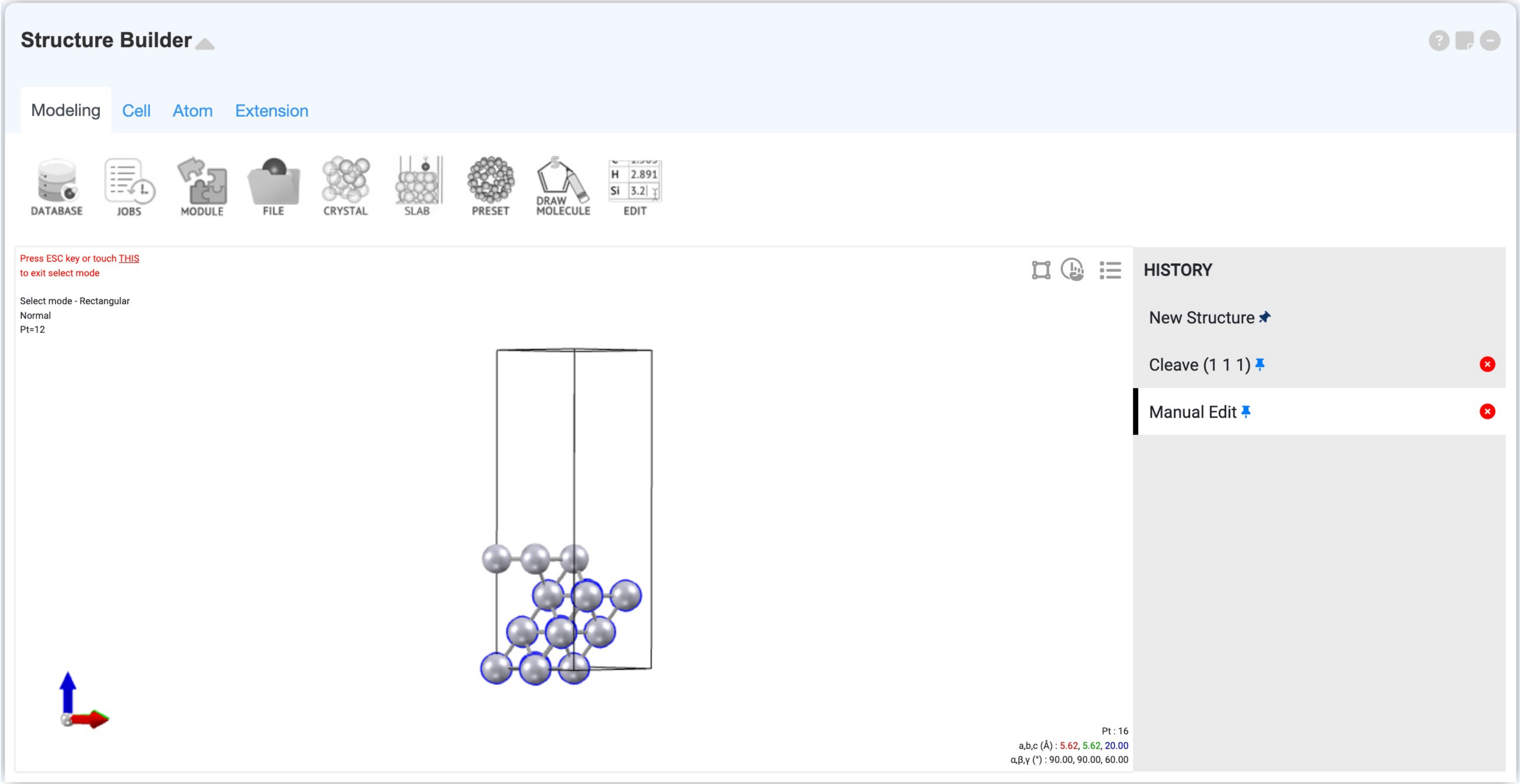


[Lab.3] Oxygen Reduction Reaction (ORR)

2. Optimize the atomic structure of Pt(111) surface



materials square

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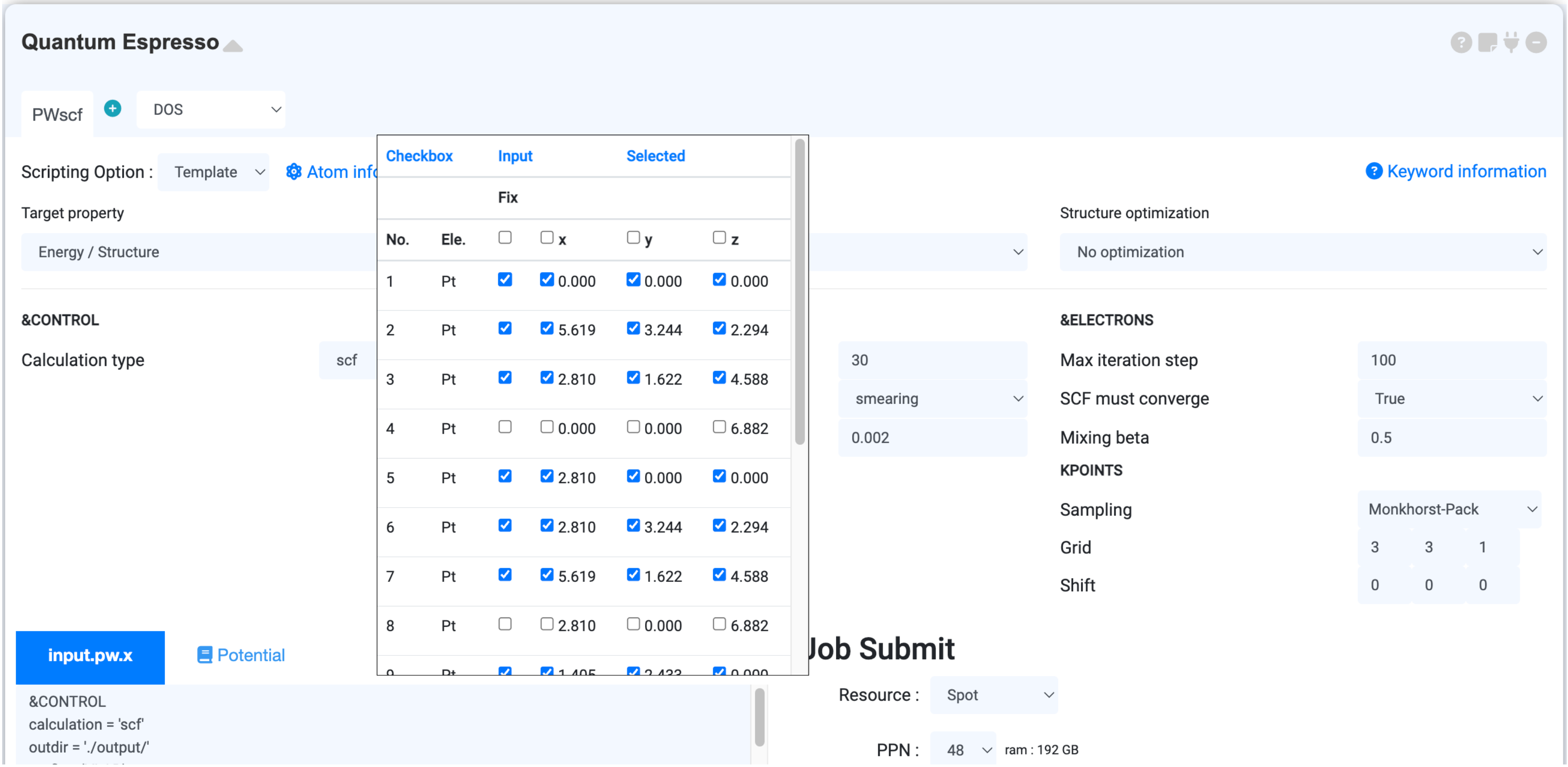

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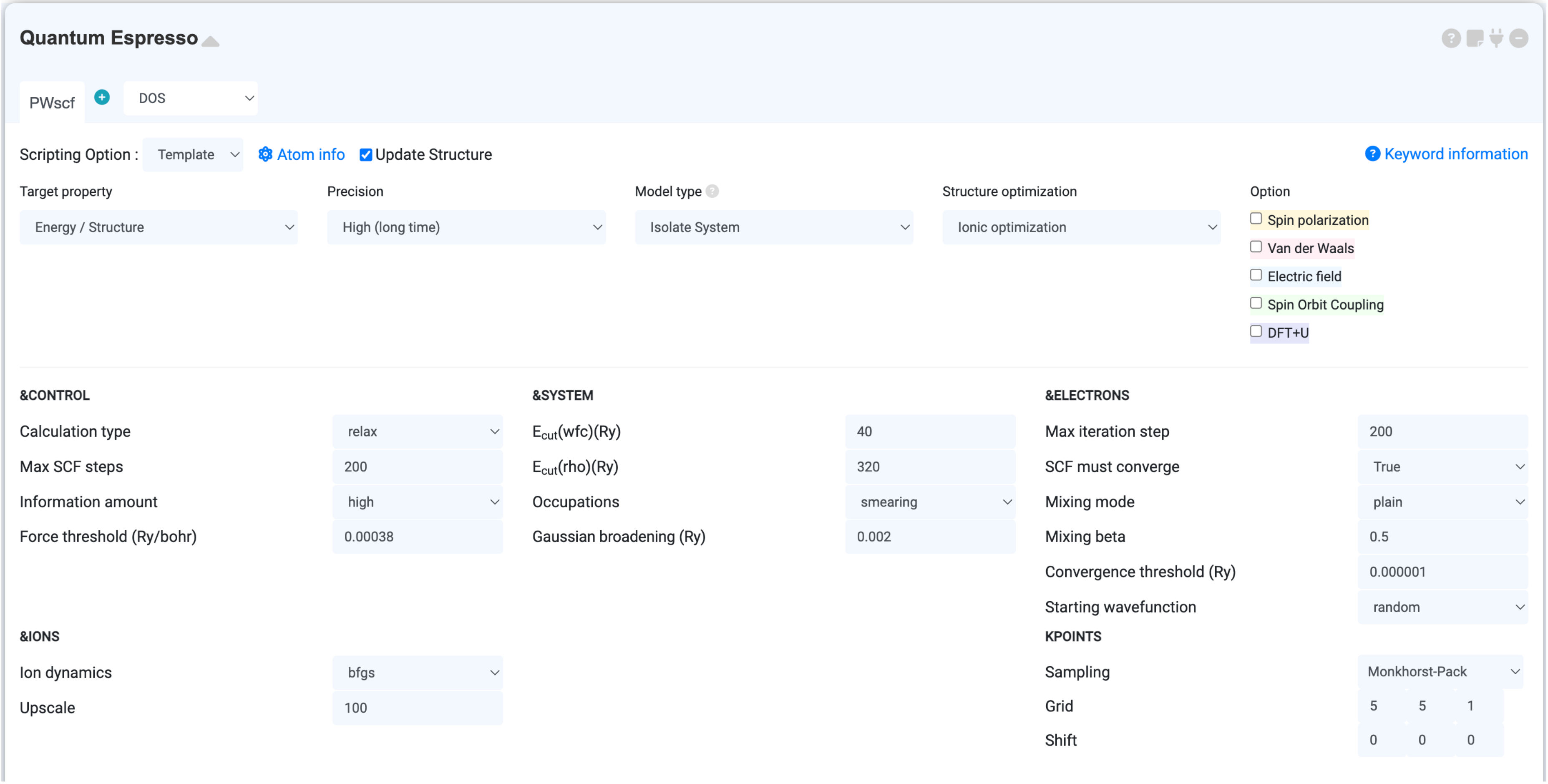
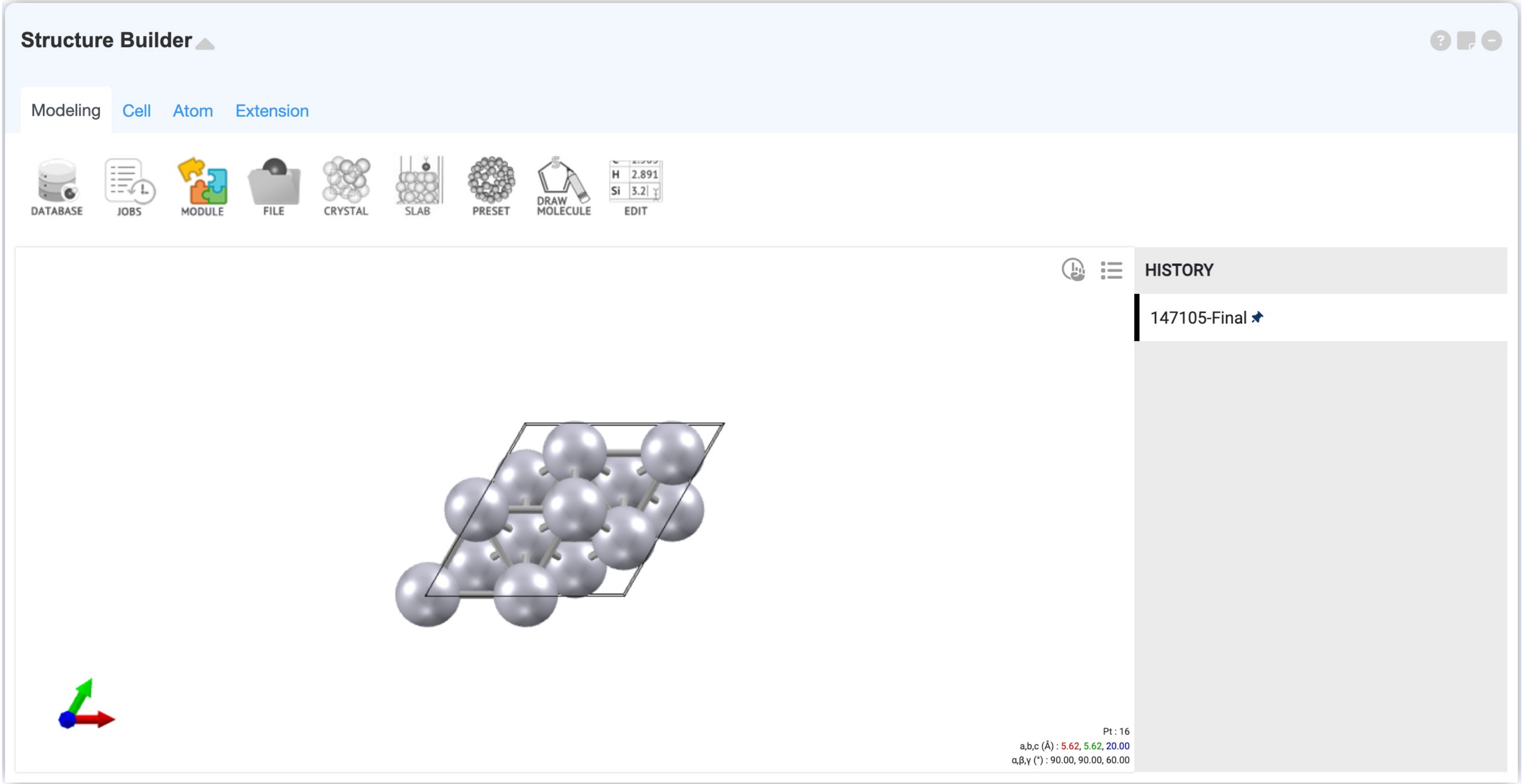


[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



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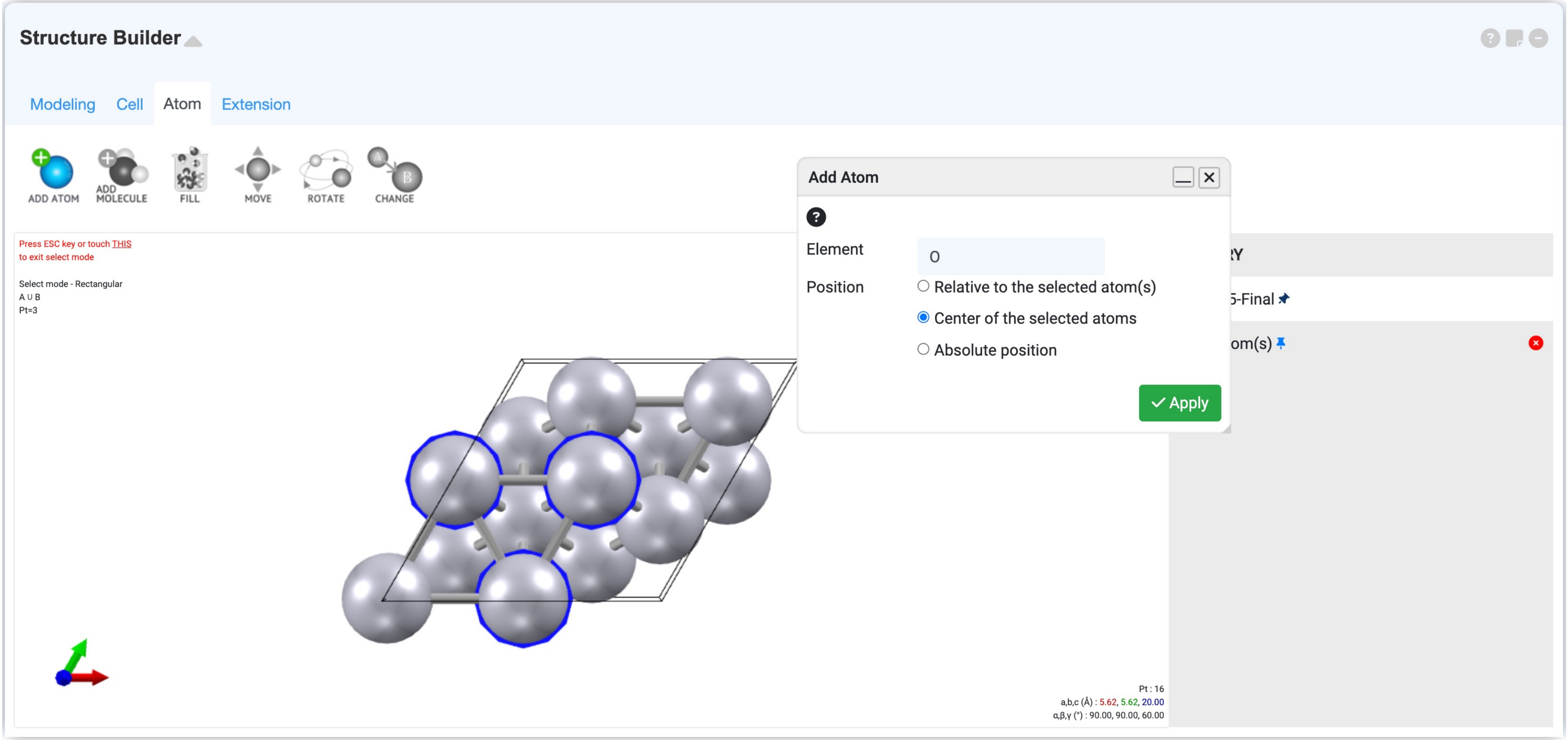

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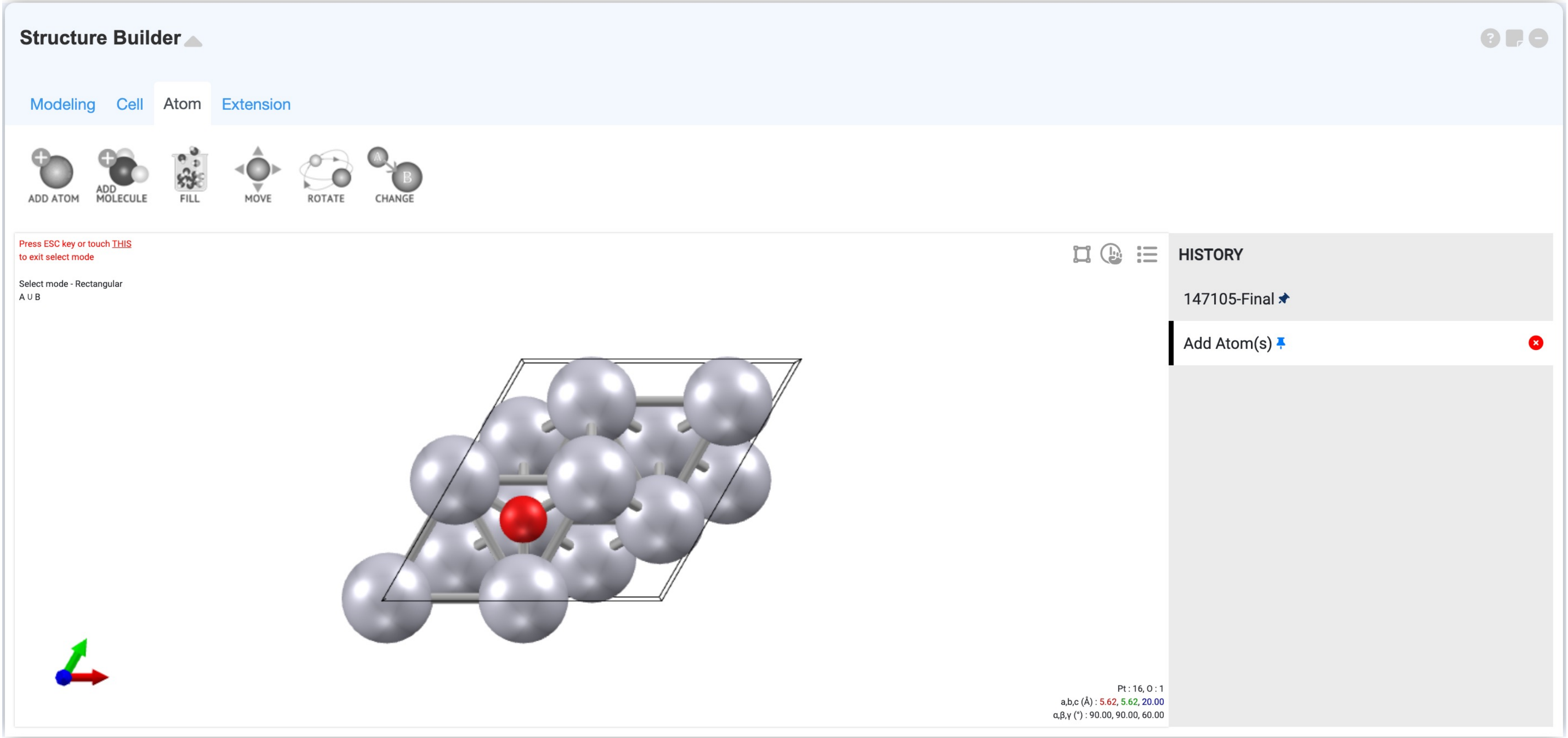

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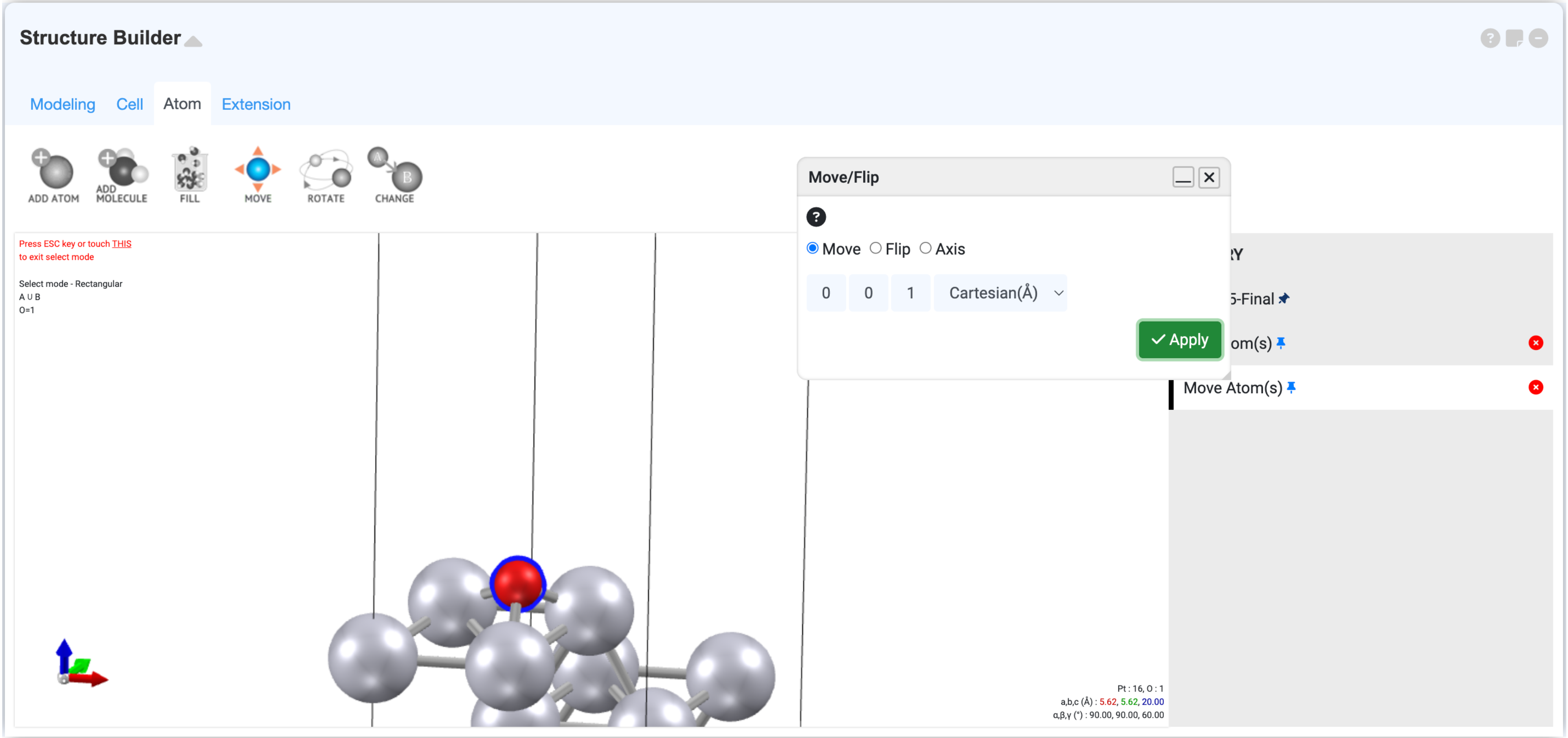

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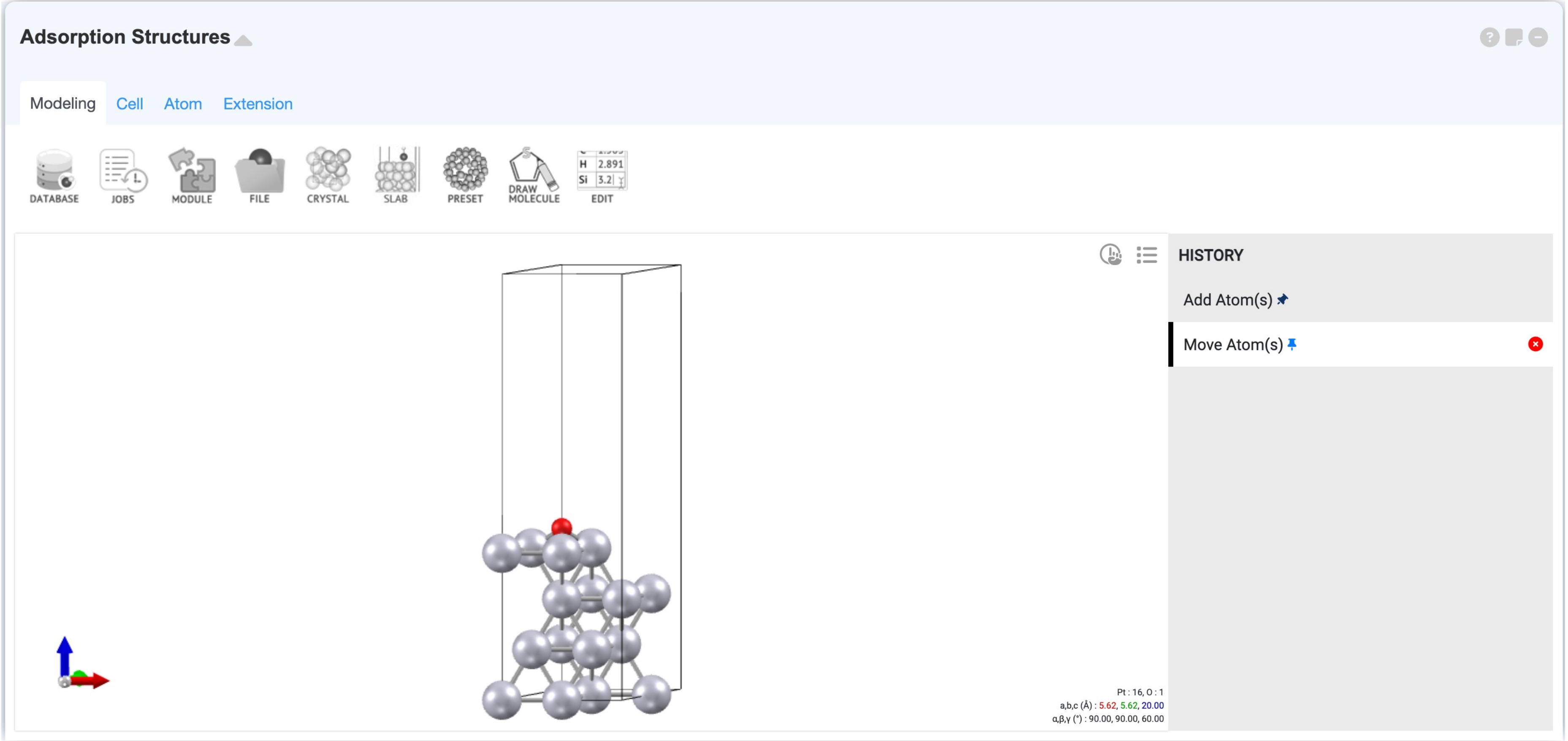

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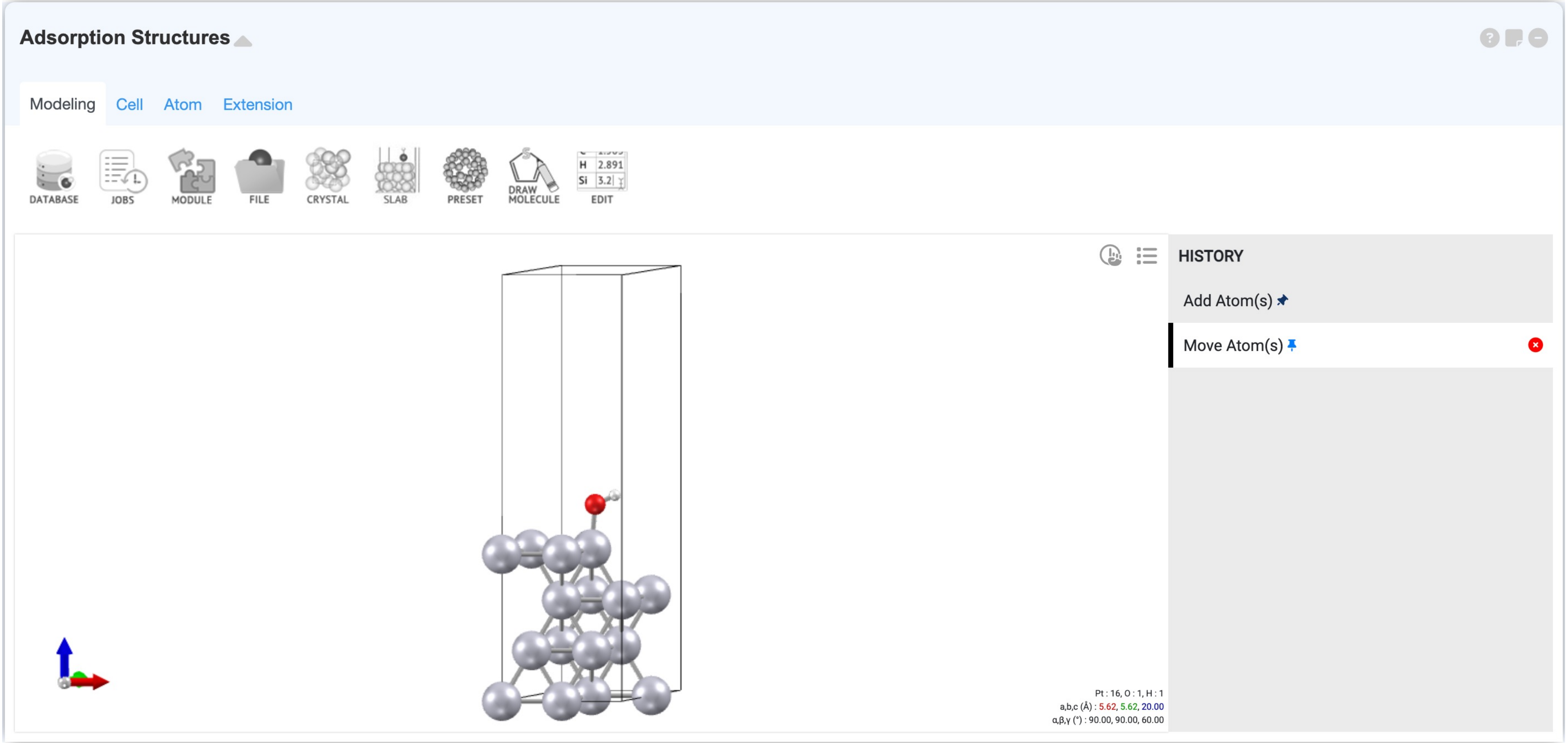

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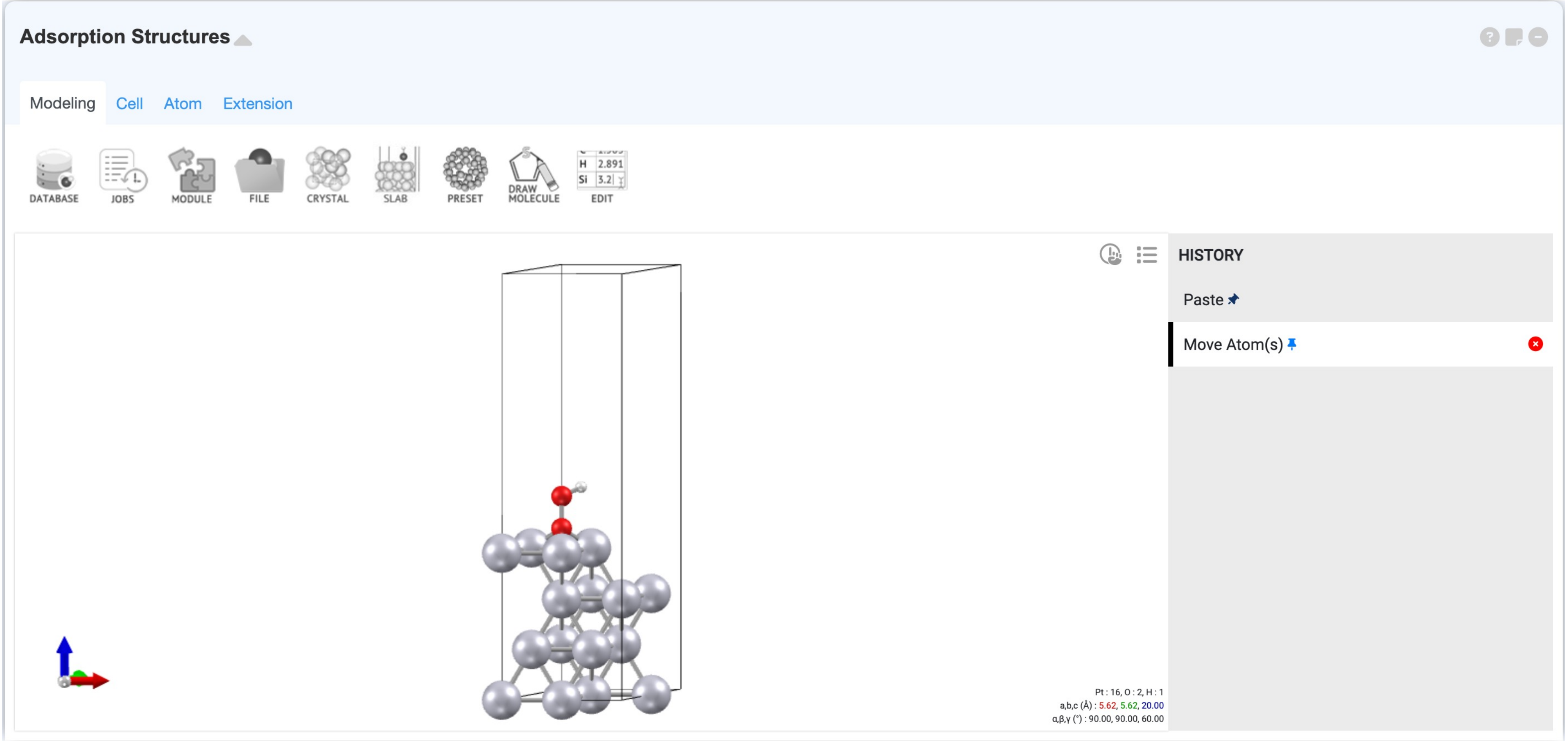

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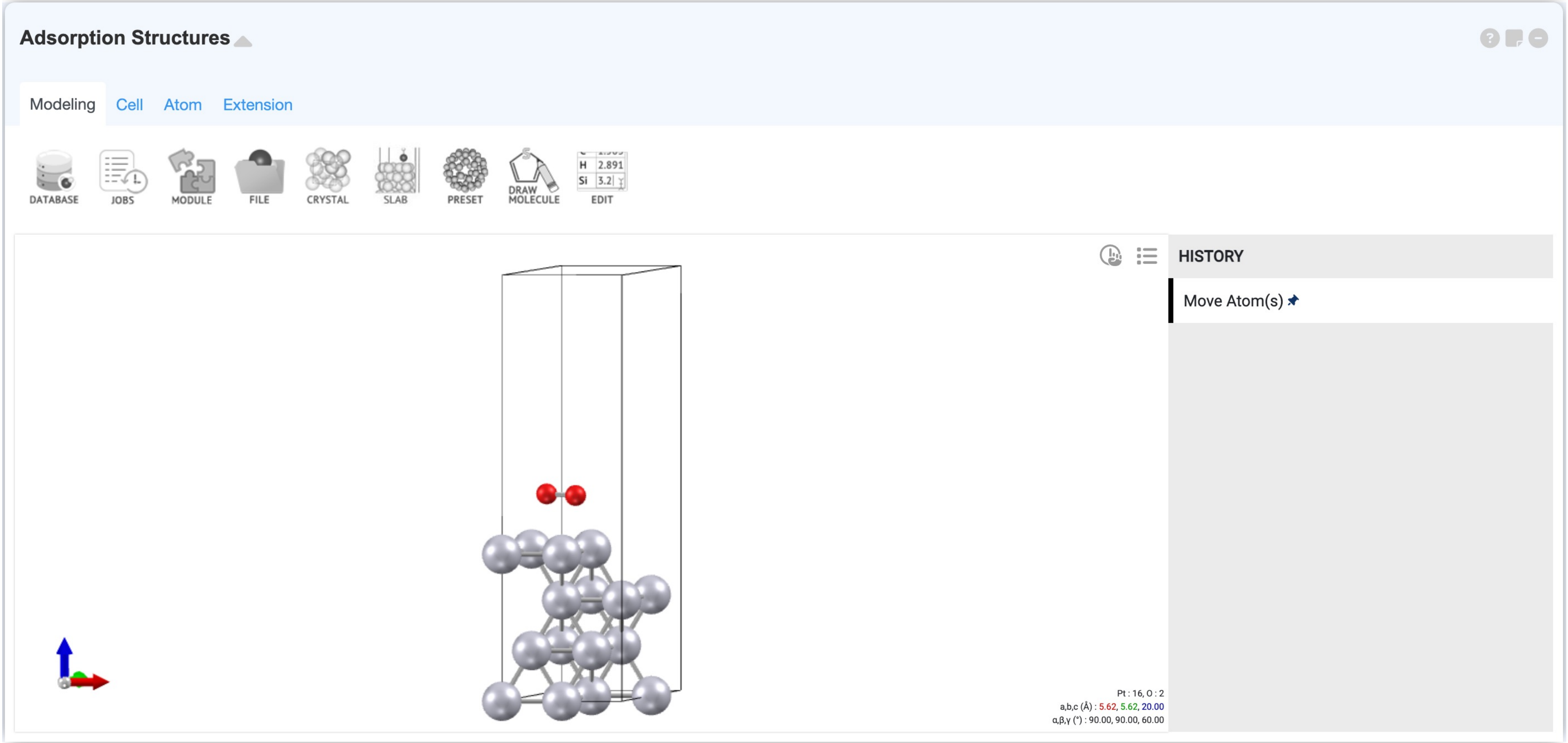

[Lab.3] Oxygen Reduction Reaction (ORR)

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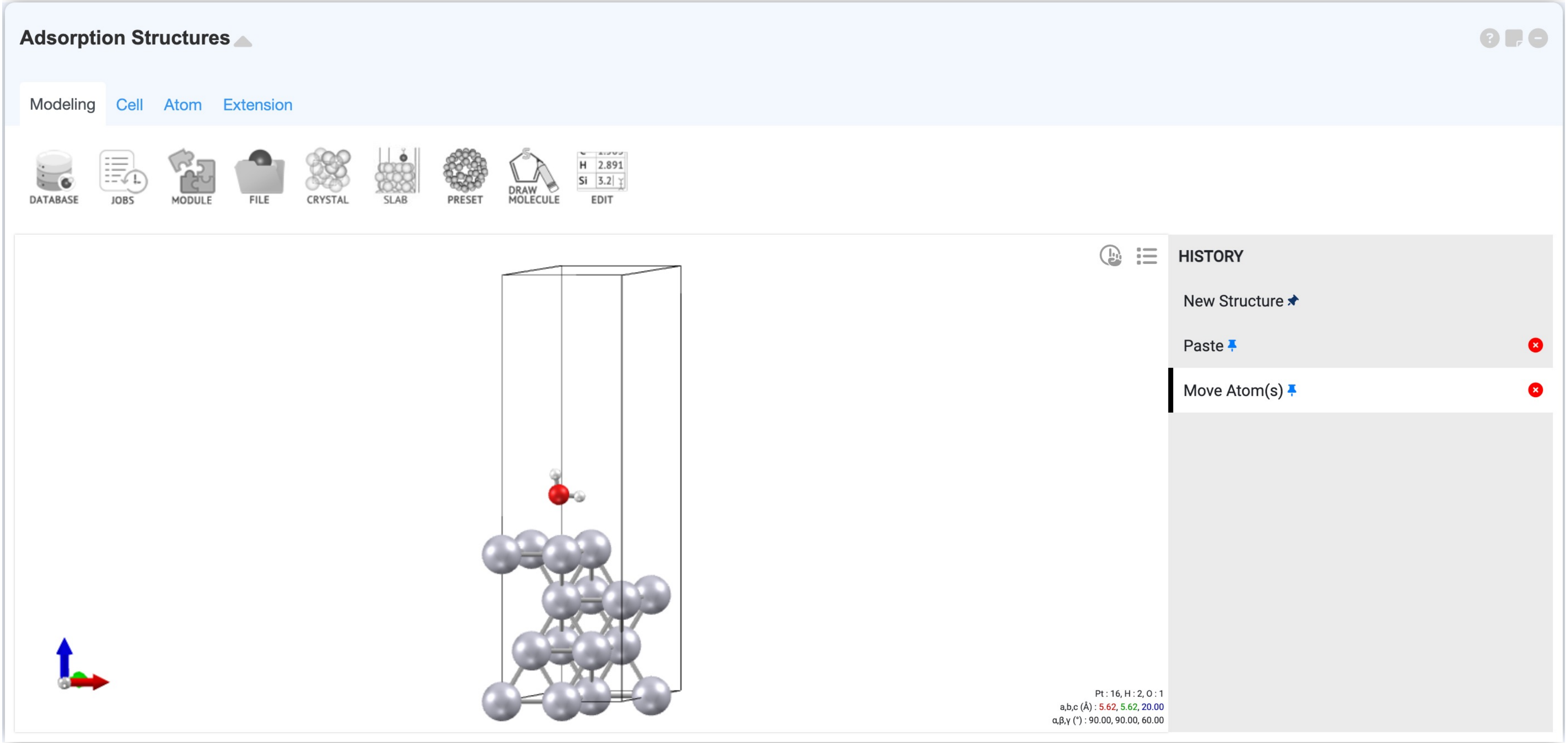
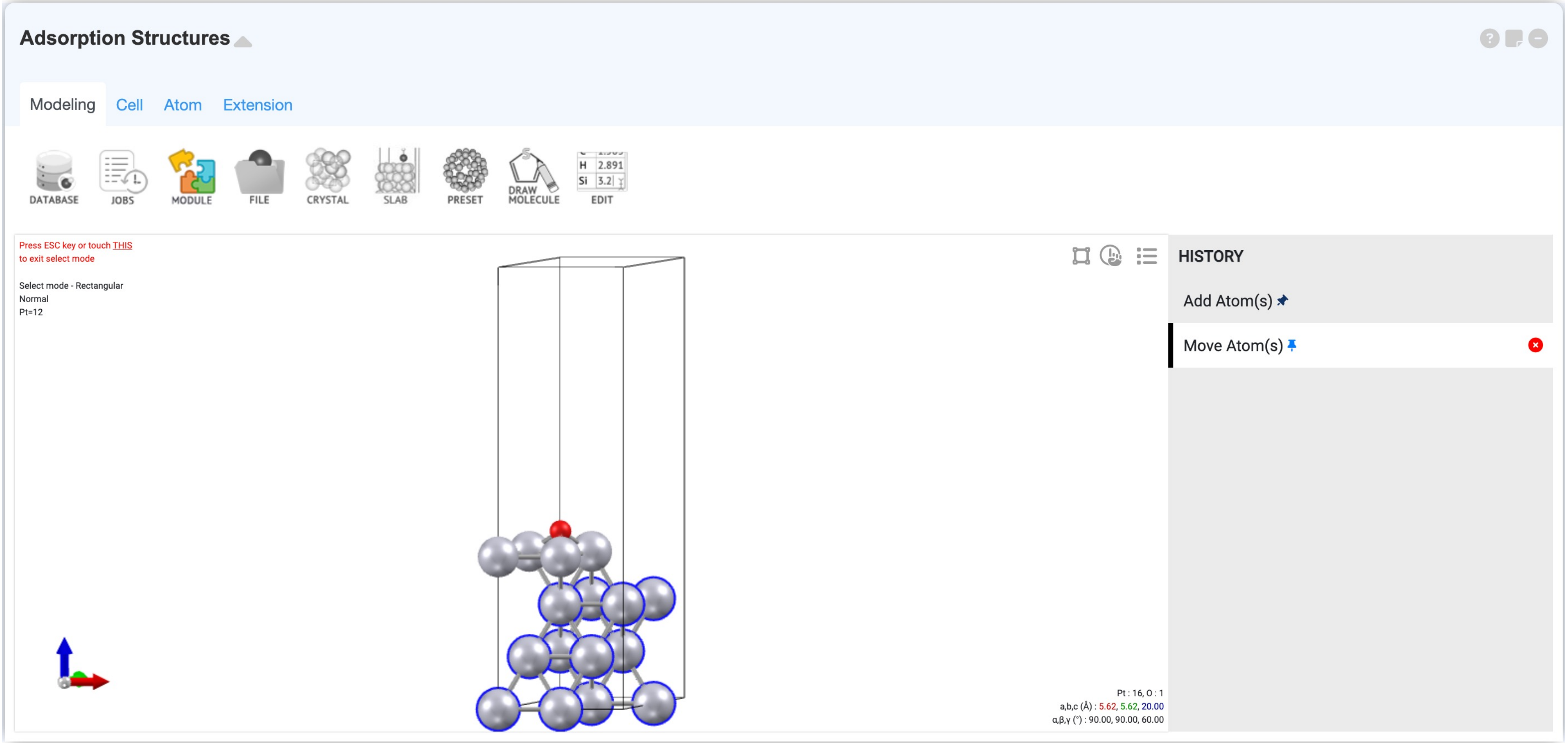


4. Optimize atomic structures (adsorption structures)



[Lab.3] Oxygen Reduction Reaction (ORR)

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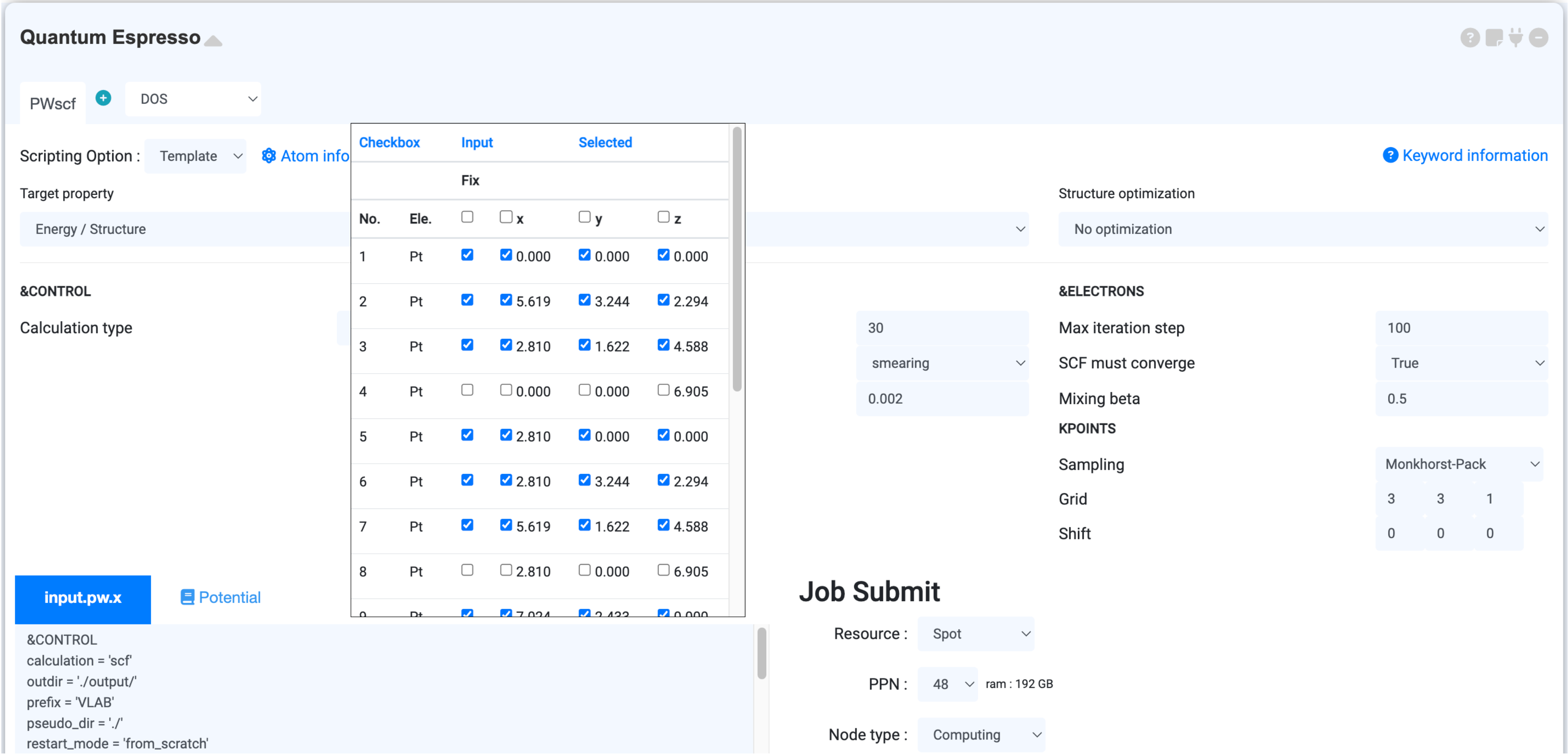

[Lab.3] Oxygen Reduction Reaction (ORR)

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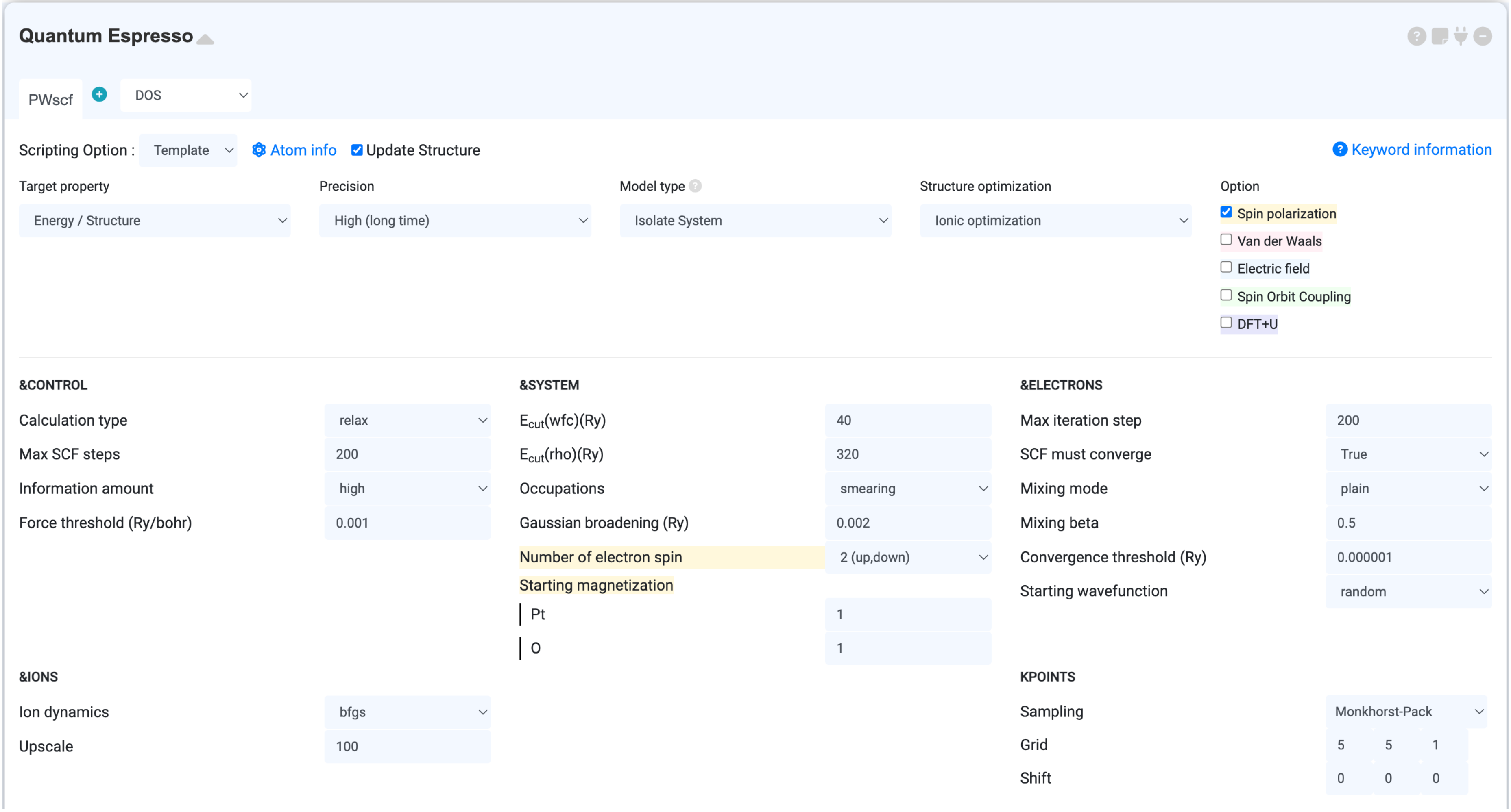
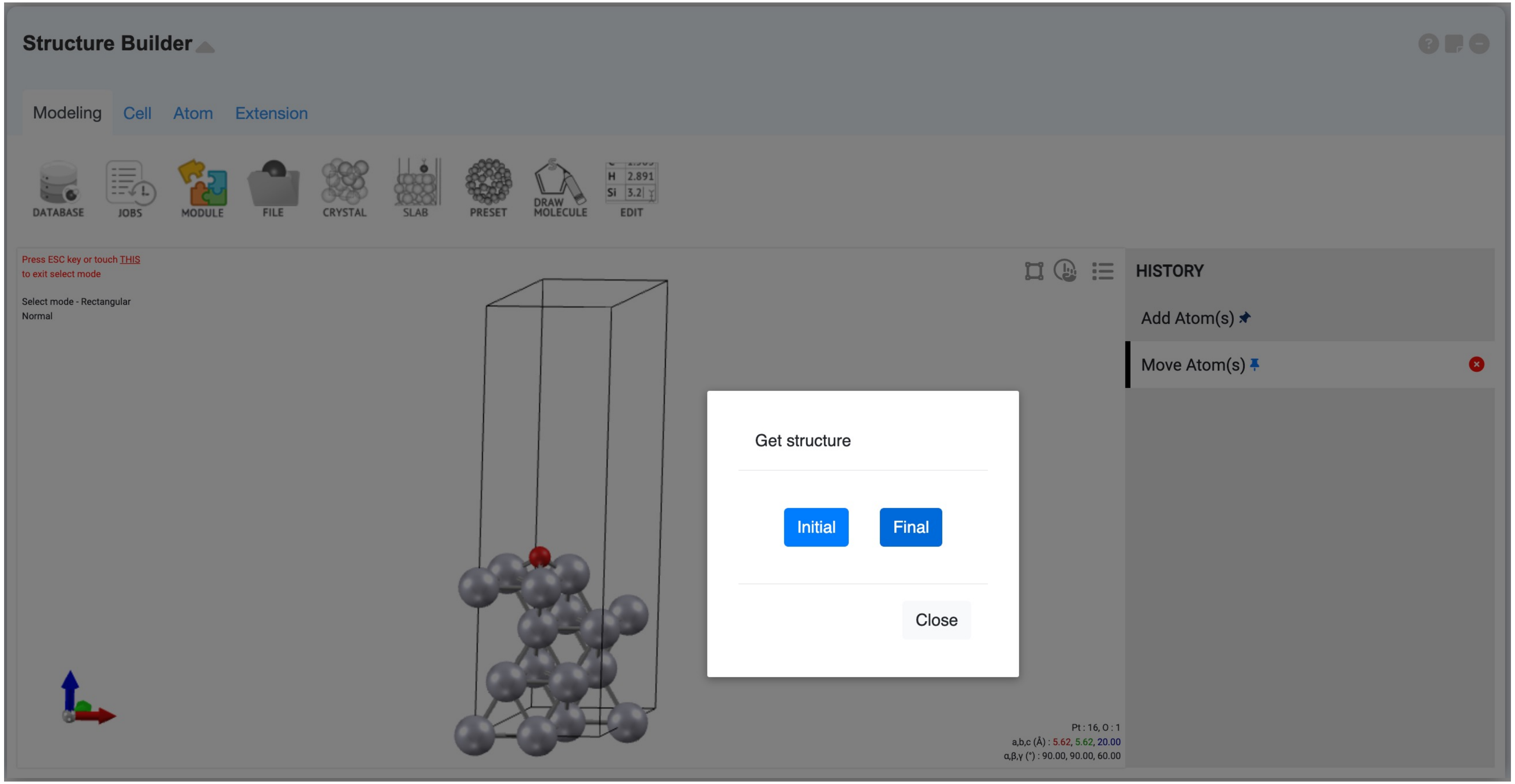


[Lab.3] Oxygen Reduction Reaction (ORR)

5. Analyze optimized adsorption structures



[Lab.3] Oxygen Reduction Reaction (ORR)

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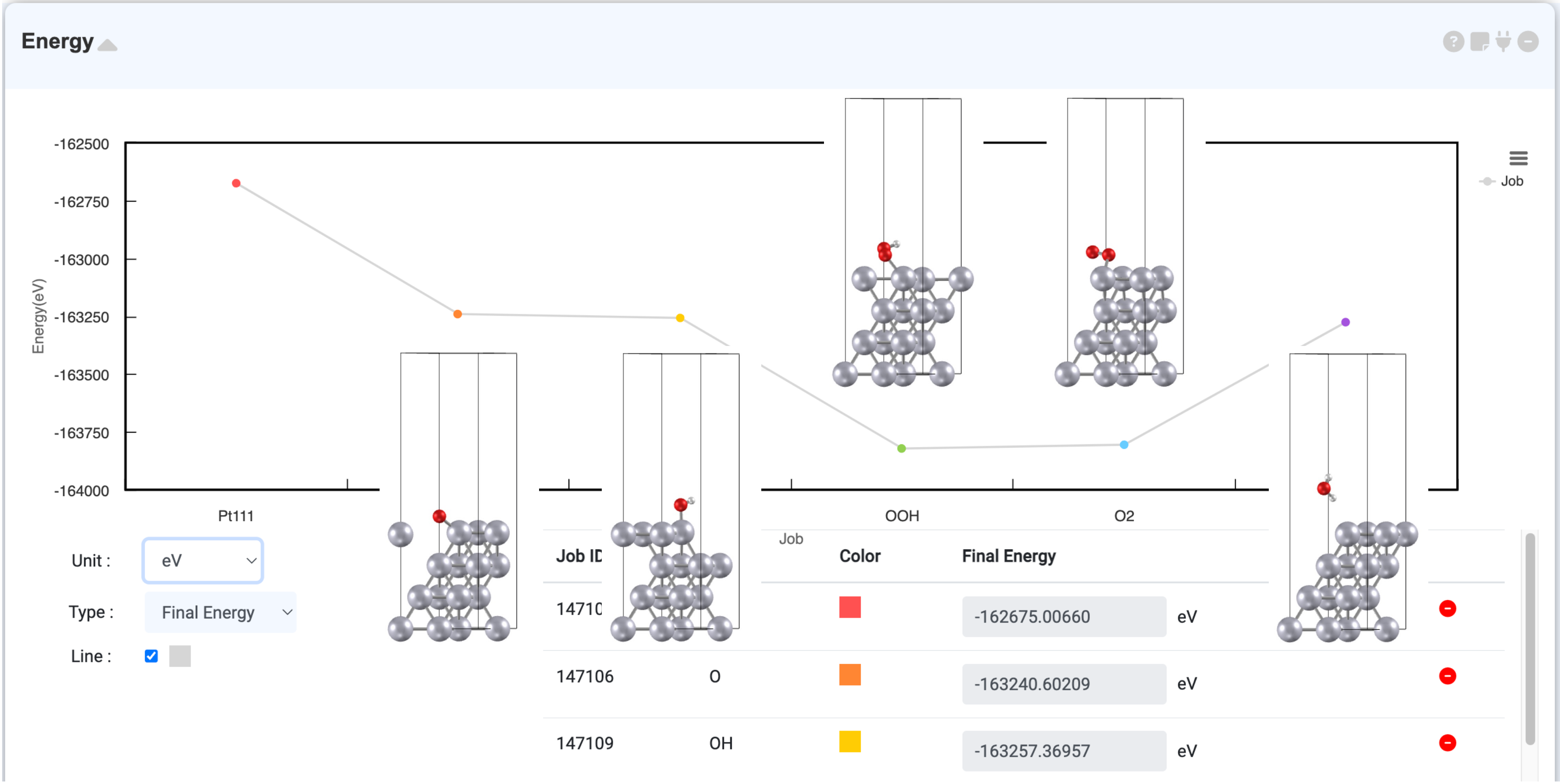
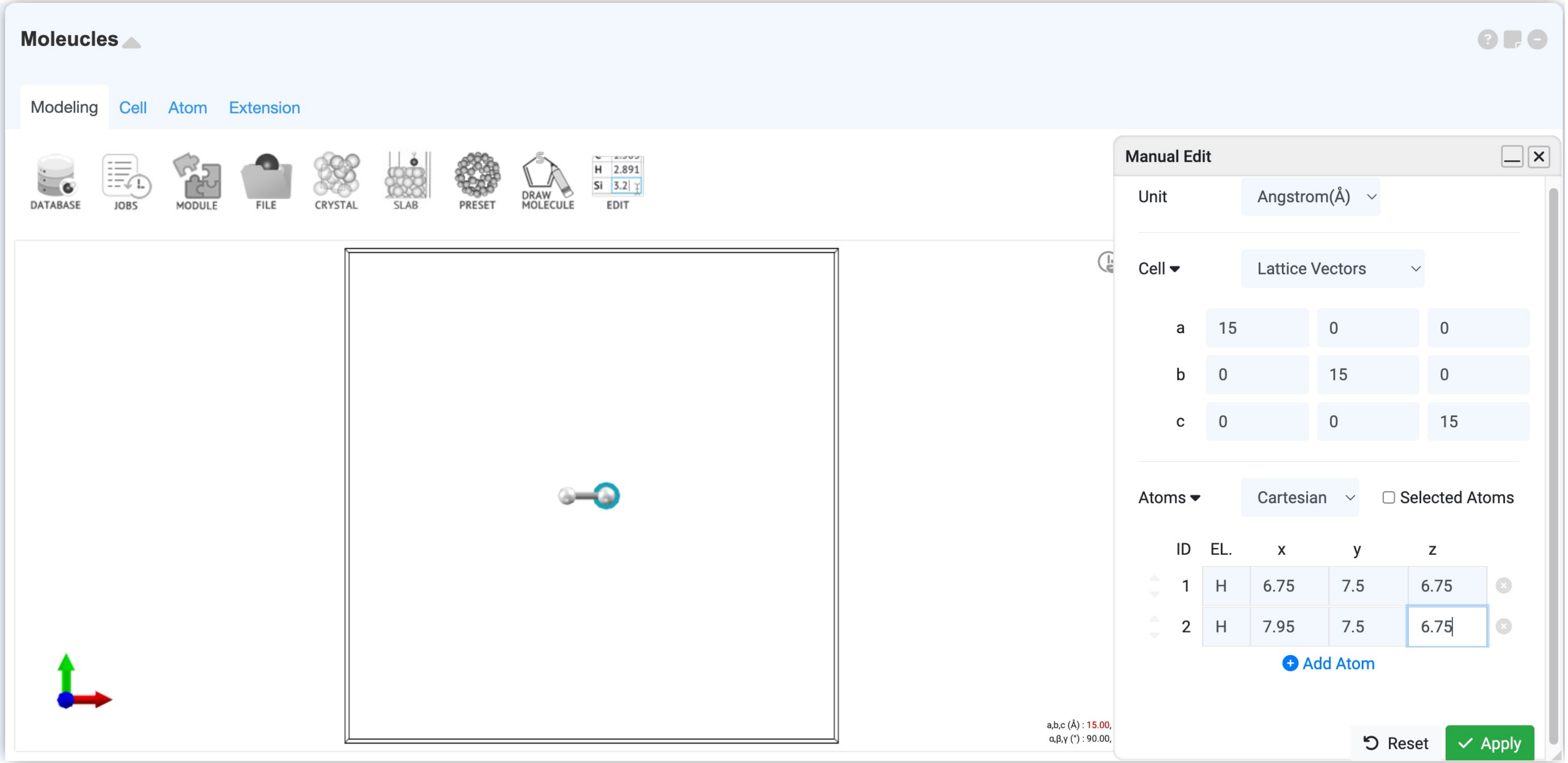
5. Analyze optimized adsorption structures



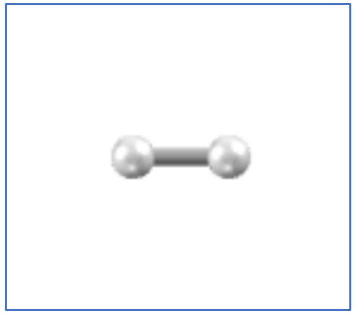
[Lab.3] Oxygen Reduction Reaction (ORR)

6. Build molecule structures

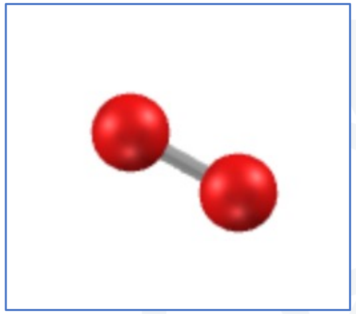


[Lab.3] Oxygen Reduction Reaction (ORR)

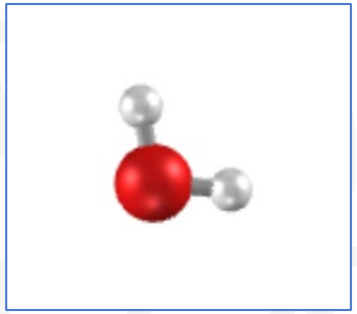
6. Build molecule structures



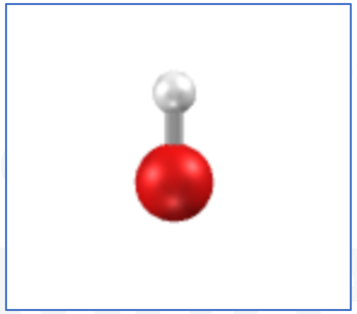
H_2



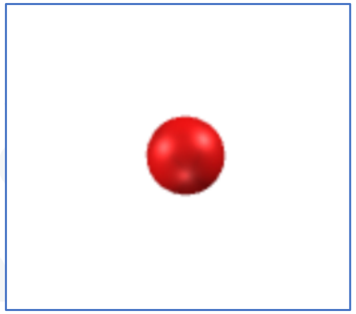
O_2



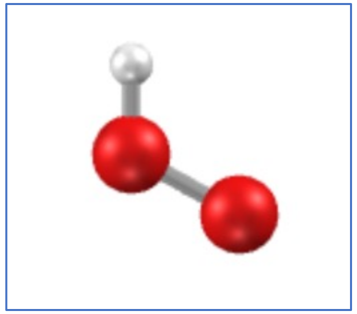
H_2O



OH



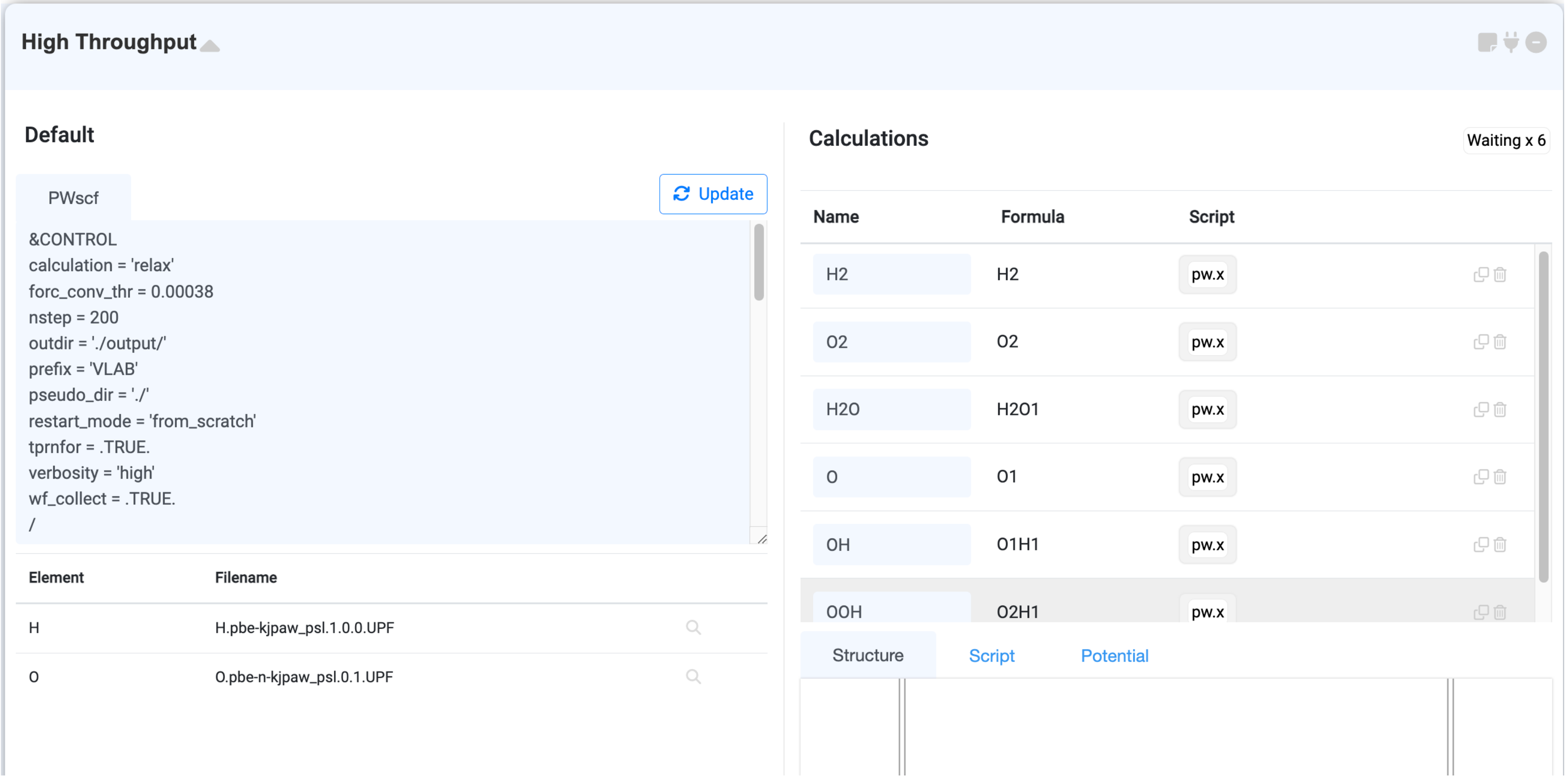
O



OOH

[Lab.3] Oxygen Reduction Reaction (ORR)

7. Optimize molecule structures using High Throughput (molecules)



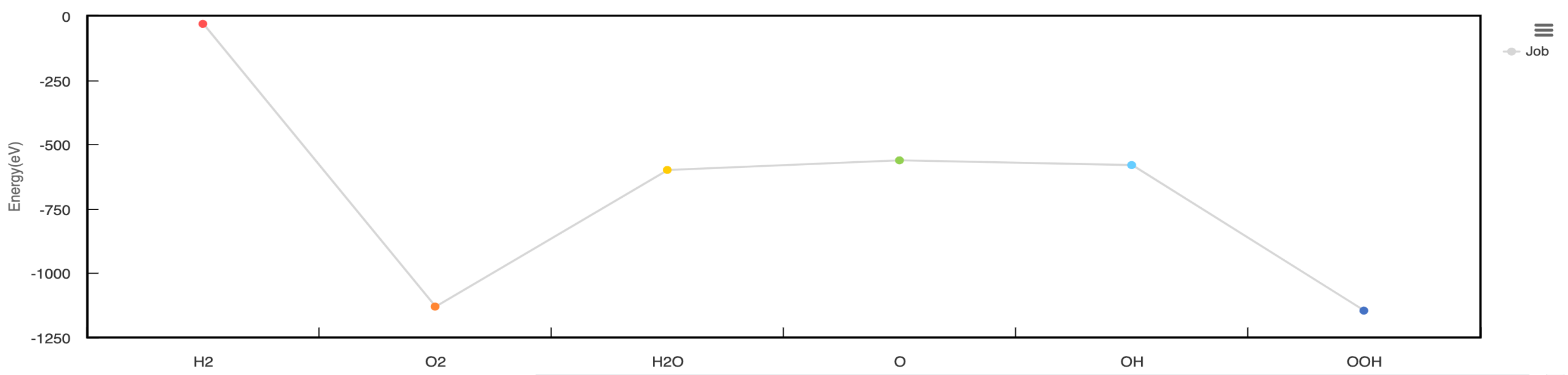
[Lab.3] Oxygen Reduction Reaction (ORR)

materials square

8. Calculate adsorption energies

$$E_{adsorption} = -[E_{adsorption\ structure} - (E_{molecule} + E_{surfaces})]$$

Energy



Unit : eV

Type : Final Energy

Line :

Job ID	Label	Color	Final Energy	
147150	H2		-31.73444	eV
147151	O2		-1129.81110	eV
147152	H2O		-599.14416	eV

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materials square

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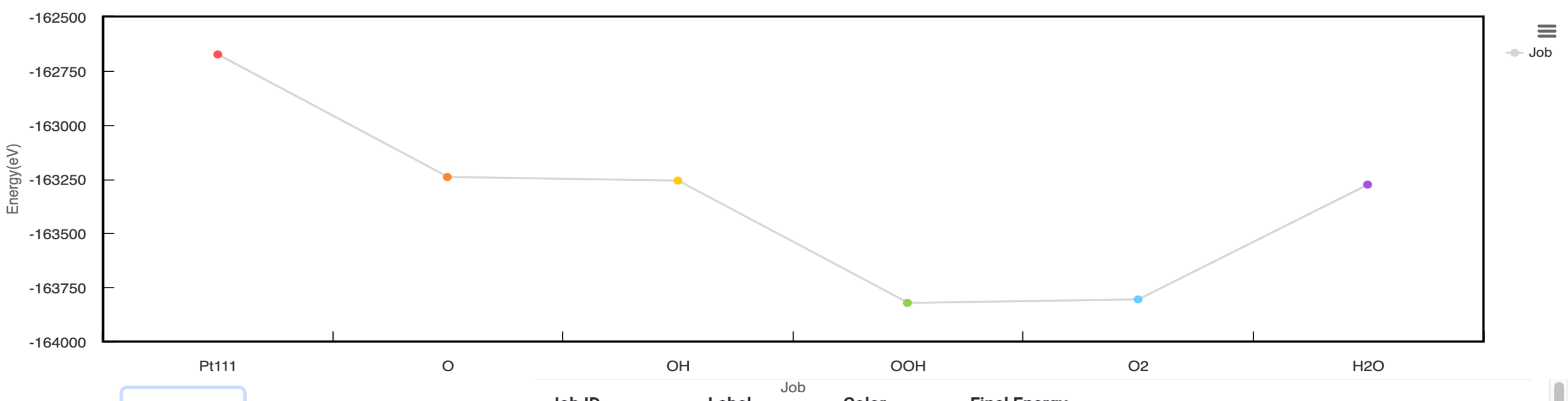


Unit : eV

Type : Final Energy

Line :

Job ID	Label	Color	Final Energy	
147105	Pt111		-162675.00660	eV
147106	O		-163240.60209	eV
147109	OH		-163257.36957	eV

[Lab.3] Oxygen Reduction Reaction (ORR)

materials square

8. Calculate adsorption energies

$$E_{adsorption} = -[E_{adsorption\ structure} - (E_{molecule} + E_{surfaces})]$$

Memo

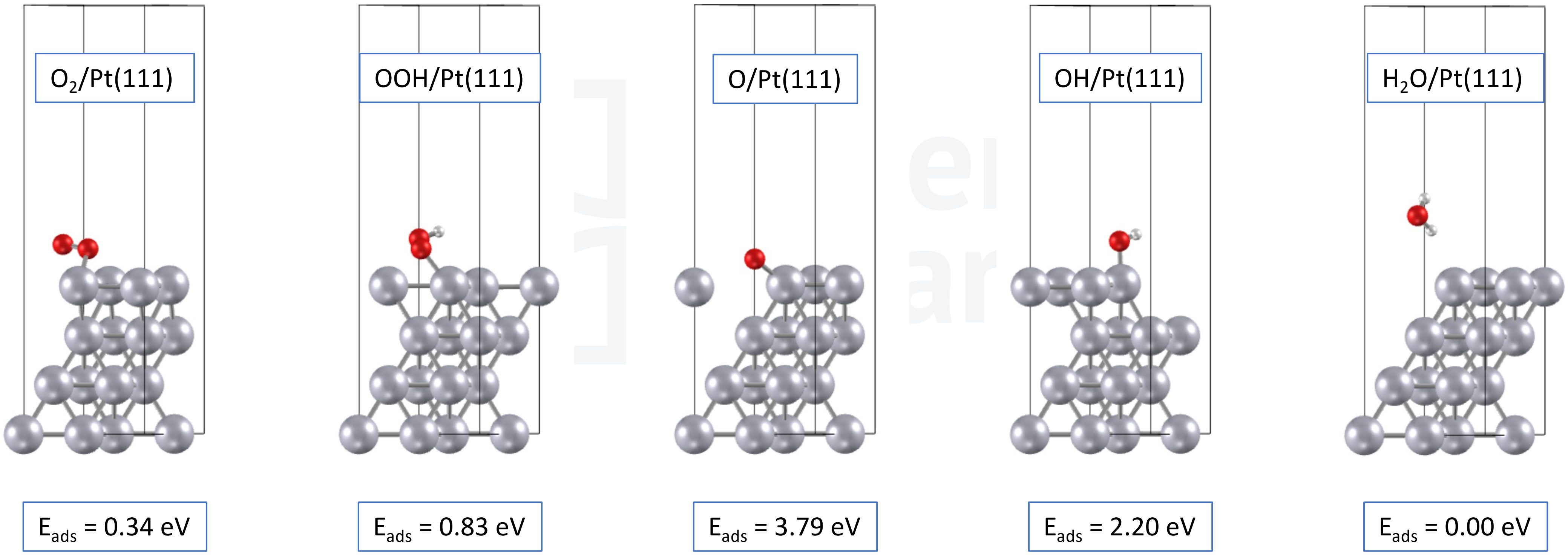
Note Sheet

	A	B	C	D	E	F	G	H	I	J
1	Structure	Total Energy			Total Energy	Adsorptior Energy				
2	H2	-31.73444								
3	H2O	-599.1441(		H2O/Pt(11	-163274.2:	0.0746199				
4	O2	-1129.811		O2/Pt(111	-163805.1!	0.3390900				
5	OOH*	-1145.653:		OOH/Pt(11	-163821.4!	0.8269900				
6	O*	-561.8097		O/Pt(111)	-163240.6(	3.7857800				
7	OH*	-580.1630!		OH/Pt(111	-163257.3(	=-(E7-(B7+B8))				
8	Pt(111)	-162675.0(								
9										
10										
11										

[Lab.3] Oxygen Reduction Reaction (ORR)

8. Calculate adsorption energies

$$E_{adsorption} = -[E_{adsorption\ structure} - (E_{molecule} + E_{surfaces})]$$



[Lab.3] Oxygen Reduction Reaction (ORR)

9. Calculate Free-energy diagram

- $\Delta E_{\mathrm{OOH}^*} = E(\mathrm{OOH}^*/slab) - E(slab) - \left\{2E(\mathrm{H_2O}) - \frac{3}{2}E(\mathrm{H_2})\right\}$
- $\Delta E_{\mathrm{O}^*} = E(\mathrm{O}^*/slab) - E(slab) - \{E(\mathrm{H_2O}) - E(\mathrm{H_2})\}$
- $\Delta E_{\mathrm{OH}^*} = E(\mathrm{OH}^*/slab) - E(slab) - \left\{E(\mathrm{H_2O}) - \frac{1}{2}E(\mathrm{H_2})\right\}$

Memo

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7	OH*	-580.1630		OH/Pt(111	-163257.3	2.1999200	=E7-B8-(B3-0.5*B2)			
8	Pt(111)	-162675.0								
9										
10										
11										

[Lab.3] Oxygen Reduction Reaction (ORR)

9. Calculate Free-energy diagram

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- $\Delta E_{\mathrm{OH}^*} = E(\mathrm{OH}^*/slab) - E(slab) - \left\{E(\mathrm{H_2O}) - \frac{1}{2}E(\mathrm{H_2})\right\}$